

Lec 47 Denoising Diffusion Probabilistic Model-Reverse Process

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Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In today's session, we will discuss about denoising diffusion probabilistic models. Particularly, the reverse process. So, in my previous lecture, we have discussed about the forward process in detail, the mathematical steps involved or mathematical formulation of the forward process. In today's session, we will discuss the mathematics of the reverse process. In detail, we will go, okay?

[00:32] So, let us start with a small recap on the forward diffusion process. What's What happens there? So, as we know in the forward diffusion process, we have a clean image to start with at t equal to zero, and then progressively we keep on adding noise, and it goes through multiple steps, and then we go to a place at t equal to capital T , where the entire image is, you know, pure noise, okay? So, this is the forward diffusion process. We all know about this, and this is the probability distribution equation for forward diffusion process.

[01:16] So, it gives the probability distribution of getting X_T conditioned on X_0 is a Gaussian distribution with this mean, square root of α_t $\bar{X}_0$, and this is the covariance term, variance or the covariance term. Now, this is the equation for the forward diffusion we know, and during inference, you know, for sampling, we get we have this equation. So, if we want to find out what is X_T , you know, we have this equation. X_T we can find out given we know X_0 . What is X_0 here? X_0 is the image at time step t equal to zero. X_T is the image at time step t equal to t small t , okay? So, these are the equations which we have already seen in the last session. And uh we have seen what is the reparameterization

[02:13] trick there and what is the advantage of this uh reparameterization? It helps us to get X_T or it helps us to sample X_T directly from X_0 without performing all the intermediate diffusion steps, okay? So, this was done in our previous um lecture, okay? Now, we will take it forward from here in today's session. So, we will look into what is this backward diffusion process, what exactly happens there, what is the probability distribution, how we can sample out uh you know uh sample one image from you know the noise, okay? So, we will see.

[02:53] So, the main idea of backward diffusion process is gradually it removes noise uh from uh you know from a pure noise it keeps on denoising till we get the clean image. So, the sequence looks like this. It starts from X_T which is the image at t equal to capital T . Then it comes to X_{T-1} and it keeps on going till X_2 X_1 X_0 , okay? So, this is the sequence for the backward diffusion process. And this is a picture which shows that one.

[03:26] X_T X_{T-1} X_2 X_1 X_0 . So, we can see that $X_{\text{capital } T}$ is complete noise and then we start denoising. It gets It starts denoising and finally we get the clean image, okay? So, it generates realistic samples by progressively removing noise from a Gaussian noise sample, okay? So, this is my Gaussian noise sample and this is my final image that we have got from here. The clean image, okay. And so the goal of the backward diffusion process, what exactly is the goal of this process? We want to learn the distribution from where we can find out X_{t-1} given X_t . So, suppose I want to find out or sample this X_{t-1} .

[04:17] What is known to me at that time? It is X_t . So, conditioned on X_t , what is the probability of getting X_{t-1} ? Okay, so this is given a noisy sample X_t , predict a slightly less noisy version X_{t-1} . So, this is the goal of the entire backward process. And here we can see this θ θ is basically the parameters of the neural network because the neural network will learn how much noise to be, you know, removed so that we can come from X_t to X_{t-1} . Because as we are going, you know, in this sequence in the backward diffusion process, we keep on denoising or keep on reducing or removing noise from the original sample or original noise sample, okay.

[05:07] Now, so the goal of DDPM training is to learn this distribution. That is the first

thing. DDPM will try to learn this distribution with the θ being the parameters of the neural network, okay. Now, DDPM assumes the reverse process is a Markov chain, okay. So, the reverse process completely follows Markov chain. And when I tell it is Markov chain, what do I mean that x_{t-1} depends only on x_t . Okay, so your current state depends only on the previous state, okay. So, that is uh your Markov chain.

[05:50] So, for Markov chain, we can write that your x_{t-1} is only dependent on x_t . Okay? Even if they have gone through all these steps to come to x_{t-1} , still the you can sample out x_{t-1} provided you know x_t . Okay? Because that is the only dependence that is there for x_{t-1} according to Markov chain. Okay? So, this is the equation for or the uh sequence for uh you know um the reverse uh process x_t to x_{t-1} then x_{t-1} to x_{t-2} until it goes to x_0 . Okay? So, there is a uh arrow missing. I'll just put that one.

[06:42] Okay? So, it goes to x_0 . Okay? Now, because of this Markov's assumption, we can write $P(\theta | x_t, x_0)$, which is a 0 to t capital T means it is the entire sequence. Okay? Is equal to $P(x_t | x_0)$. What is this x_t ? x_t is your starting point. Okay? Starting point of your backward process. Into the product of all these conditional probabilities. Okay? Conditional probability of x_{t-1} given x_t . Okay? So, because of Markov's assumption, my $P(\theta | x_t, x_0)$ can be written in this form. So, we will look into what exactly they mean. This basically gives you the joint probability distribution for the entire reverse trajectory.

[07:35] And this $P(x_t)$, this is the prior distribution over x_t . That means we are starting from x capital T . Okay? So, this x capital T is the starting point of generation and we call it as the prior distribution. We know only x_T . x_T is known to us x capital T . Okay? And then this is the product of all these you know, probability values conditional probability values that we get. Okay? And θ is the parameters of the neural network. Okay?

[08:09] So, what we have seen here is the reverse back or the reverse process can be written as a joint probability distribution and that is equal to the prior probability distribution of the reverse process starting from x_T into this the product of all the conditional probabilities. Okay? And what is my x_T ? x_T is Gaussian noise with zero mean and unit variance. Okay? Now again, we will come back to forward diffusion process equations. Now, the a question may arise that why we are going back and forth from forward to you know, forward process and backward process. We are studying about the reverse process or the backward process in this lecture.

[08:56] Why we are going back to the forward process? The The reason we are going back is our backward learning process is completely you know, guided by the forward process because in the forward process we it it was a known process where we have added noise in each and every step with increasing amount of noise getting added in

the image and this and this is the process which will help the neural network to learn the reversing or the denoising process. Okay? So, that is the reason we are going back to the forward equations of the forward diffusion process as well.

[09:35] Okay? So, this we have seen uh you know uh in my previous slide also, X_T given X_0 is base is a Gaussian uh distribution with this is the mean. So, this mean this this term is the mean and this is the covariance or the variance term, okay? Now, this is the probability distribution of forward diffusion process. We all know this one. This is a known thing which we have done in our previous lecture also. Now, before I move forward, I would like to introduce you all with two terms. One is the prior distribution and one is posterior distribution, okay?

[10:16] Because this is very important in uh when we are discussing about denoising diffusion probabilistic models, okay? So, what is this prior distribution? The probability distribution that represents what we know or assume about a variable before observing any additional information. That means the distribution that we are assuming before getting some evidences, that is called as the prior distribution, okay? Whereas posterior distribution is a probability distribution after observing additional information. You have seen some you have got some evidences, you have seen some observations, and from there you are getting a distribution which is your posterior distribution, okay?

[11:02] We will use this term posterior distribution in my um coming slides, that's the reason I have um discussed about it. So, now we will look into the posterior of the forward process. What exactly is the posterior distribution for the forward process? The posterior distribution looks like this, $Q_{X_{T-1}}^{X_T, X_0}$. So, your X_{T-1} is now conditioned on X_T and X_0 . So, under forward noising process Q What is the probability distribute What exactly uh Okay, under the forward noising process Q , what is the probability distribution of the previous step? So, this is what is the probability distribution of previous step X_{T-1} given the current noisy state is X_T and you know X_0 .

[11:54] Okay. So, what exactly the posterior is giving me? When we are going at the forward noising process What is the probability distribution for X_{T-1} ? So, X_{T-1} is the image at time step small $T-1$. Okay, it's a noisy version. Uh it is also a noisy version, but X_T What I have in my hand is I have seen X_T and I know X_0 . Okay. So, under the forward So, this is called as the posterior of the forward process. So, what exactly it means under the forward noising process Q , what is the probability distribution of the previous step?

[12:40] So, X_{T-1} is my previous step because we start in the forward process we start from X_0 , X_1 , X_2 dot dot dot we go to X_{T-1} , then we go to X_T . Okay, and then we go to $X_{capital T}$. Okay. So, here we know the starting point which is my X_0 , which is my clear image. Now, probability distribution for X_{T-1} given I know this X_T . Okay,

so this is called as the posterior. Now, this posterior of the forward process can be written as a Gaussian with a μ with a mean. This is my mean term, and this is my covariance term, okay? Covariance or the variance term. Now, uh what exactly it means? So, my x_{t-1} can be sampled if I know the mean μ_t , which depends on x_t and x_0 , and the variance term, okay? So, this is

[13:52] the posterior of the forward process. So, first we have seen how the posterior equation looks like, the posterior probability distribution looks like, and then we are telling that it can be modeled as a Gaussian distribution with this mean. This is my mean, and this is my covariance. Okay? Now, this mean I'm not I have not We are not going to the derivation of this mean, but the mean μ_t , which depends on x_t and x_0 , can be written as this equation.

[14:35] For time being, we will just see this is the equation. We are not going into the derivation of this equation, but x_t this is dependent this μ_t is dependent or the mean is dependent on x_t and x_0 , and it is equal to square root of α_t times μ_0 plus square root of $1 - \alpha_t$ times x_0 . What is my μ_0 ? μ_0 is This we have discussed in our previous lecture also. It's the product of α_s , where s equal to one to t , okay? And β_t times x_0 plus α_t times x_t . So, and this is the term for the variance, okay? Now, so this is a big equation, okay? So, once I know the mean and the variance, then I can find out what is my x_{t-1} because we know x_t and x_0 can be

[15:39] written in terms of the mean and the standard deviation, okay? So, that we know. So, this is a Gaussian which can be computed, okay? So, there is no problem because once we model our distribution as a Gaussian distribution, it becomes easy for any sort of calculations, okay? Now, coming to the reverse distribution, you know? So, we know in the reverse distribution, we start from x_t , then we come to x_{t-1} , and then x_{t-2} , x_{t-3} , x_{t-4} , like this we come, okay?

[16:15] Now, this can be written as if I want to find the probability of x_{t-1} conditioned on x_t , it can be written this is the probability of x_{t-1} given x_t , it is written as p_θ because θ is the parameters of the neural network, and why always in the reverse process we are putting θ because it is the learning process. That is where the neural network is learning. What exactly it is learning? It is learning the probability distribution of x_{t-1} conditioned on x_t , okay?

[16:53] Because when we are coming in the reverse process, x_t we have, x_{t-1} is the one which is coming, okay? » [snorts] » This can be modeled as a Gaussian distribution with with this is the mean and this is the covariance term. What do we see in here is my μ_θ , which is the mean that my neural network will learn. Wherever there is a θ , that means there is some learning. The neural network will learn this one, okay? So, μ_θ depends on x_t and t . Similarly, your covariance term uh this one, Σ_θ , is also dependent on x_t and t .

[17:40] That means the reverse diffusion process in the reverse diffusion process, the probability of x_{t-1} given x_t can be modeled as a Gaussian distribution with this mean and this variance, okay? So, this is my mean. This is my covariance. Okay? And we can sample x_{t-1} from this distribution with mean $\mu_{\theta}(x_t, t)$ and covariance $\Sigma_{\theta}(x_t, t)$, okay? So, once we have this Gaussian distribution or the probability distribution of the reverse process modeled as a Gaussian with μ with μ_{θ} as the mean and Σ_{θ} as the covariance, we can write x_{t-1} , you know, from μ_{θ} and Σ_{θ} , okay?

[18:49] So, the next denoised image x_{t-1} is sampled from a Gaussian whose mean and variance are predicted from x_t . So, I have x_t , I can predict x_{t-1} , okay? So, not predict, I can Yeah, I can sample x_{t-1} from the Gaussian whose mean and variance depends on x_t . x_t we have we know. So, this is First, we have x_t , then we have x_{t-1} . Okay? So, like this. So, So, the neural network must learn the mean and the covariance. So, if I want to know from my x_t , I'm coming back like this, $x_t, x_{t-1}, \dots, x_1, x_0$, and this goes to x_1, x_0 , like this.

[19:43] Then, I can get x_{t-1} if I know the mean and the covariance term, which is dependent on x_t . Okay? So, that is the crux of the story in this slide. Then, we will Once again, we will recall the mean and the variance equation from my previous slide. So, this is the equation for mean, and this is the equation for variance. Okay? We have seen. So, big equation. Now, the thing that I want you to have an attention is there is a term x_0 here.

[20:23] What is x_0 ? x_0 is my clean image. That means, if I want to find out μ_t , okay? Then, I need to know x_0 . Because there is a term x_0 here. Can we use this equation for finding out the mean? Can we? The simple answer is no, because the posterior mean depends on x_0 , which is unknown during the generation. During generation, that means during the backward diffusion process, we are We are coming from $x_t, x_{t-1}, \dots$ as I wrote in the previous slide also, $x_t, x_{t-1}, \dots, x_1, x_0$. Okay, so we have started our generation from here.

[21:23] Generation starts here. And if I want to know this one, x_{t-1} , I know till this. I don't have an access to this. So, how do I find out the mean and the variance, which depends on x_0 . So, this equation directly we cannot use, because it has dependency on x_0 , which is my clean image. What is the solution? The solution to this one is, can we rewrite these equations without x_0 ? That means, you have to substitute x_0 with something else, and then we can see if we can find out the mean and the covariance, okay?

[22:14] So, from the forward diffusion process, we know this is the sampling equation, $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$, uh you know,

epsilon, okay? So, this equation we have already done in my previous session, and if we can make from this equation, if we can make x_0 as the subject of the formula, then we can write it as x_0 equal to x_t minus square root of one minus α_t bar divided by square root of α_t bar into epsilon, okay? So, this if we can do, that means we are writing x_0 in terms of x_t .

[22:55] Okay, now if we use this equation for my μ_t , I get this equation. Once again, there are some mathematical steps in between and which we are not showing. I'm just using this equation. This is the equation for finding out the mean, where the mean only depends on x_t . There is no dependence on x_0 because I have substituted x_0 with you know, I have written x_0 in terms of x_t from this forward diffusion process equation, okay?

[23:33] And if I replace the true noise by the noise predicted by the neural network, the equation looks like this that. So, μ_t , this depends on x_t t is what equal to 1 by square root of α_t bracket x_t minus β_t divided by square root of 1 minus α_t bar epsilon uh you know, a function of x_t {comma} t , which depends on θ . So, θ is the parameter of the neural network here, okay? So, what exactly we have done? These are all mathematical steps. What do we have learned here is we want to find out what will be the image in the in the reverse process? We want We are generating images, okay? And we are starting our generation of images from a very noisy image, which is your x_t .

[24:32] Now, then you are coming one step one step one step by denoising that image and trying to come to a place where you get the complete clear image. In the process, what you are trying to do is you are modeling the probability distribution of getting x_{t-1} given x_t as a Gaussian distribution with certain mean and covariance. Once we know that mean and covariance, once the neural network learns the mean and the covariance of that Gaussian distribution, we can sample x_{t-1} from x_t .

[25:13] And that is the reason we have done gone through all these mathematical steps, and there is an equation which we got, which gives μ_t , that means which gives the mean learned by the neural network. It depends on x_t , and there is no dependence on x_0 . Okay? So, we I don't need to know the other steps which are, you know, coming after x_t . So, that is the whole story here. Okay? So, now we have written $p(\theta | x_{t-1})$ conditioned on x_t is a Gaussian with this mean and this variance. Okay? So, this is once again I'm writing this mean, and this mean is dependent on x_t and t , and this is my covariance term or the variance term.

[26:13] Or variance term. Okay? Now, so the sampling equation, as we all know, when we have a Gaussian distribution with a certain mean and a you know, variance, we can sample x_{t-1} , which can be written as $\mu + \sigma \epsilon$. What is your σ ? That is coming from your variance. Basically, it is the standard deviation.

Okay? So, μ plus standard deviation, that into Z . Z is your um, standard Gaussian with zero mean and unit variance.

[26:54] This can be written x_{T-1} can be written in terms of the mean which depends on x_T and T and the standard deviation, okay? Multiplied with Z . So, x_{T-1} , I have substituted this μ_θ from my previous equation and this is the equation that I get. That means I am a or the neural network now has learned how to find out the next image in the reverse trajectory x_{T-1} given it knows x_T . What it has done, it has learned the mean and the covariance term in the Gaussian distribution and using that one it has sampled x_{T-1} or generated x_{T-1} , okay?

[27:43] So, this is basically your uh, you know, reverse diffusion process equation. And then we come to the training of the diffusion model and the loss function. So, the loss function of the diffusion model is given by it is what exactly is a diffusion model predicting? It is predicting the amount of noise that is there in a particular image. So, this is basically what happens is the loss function can be written as L which is expectation value over x_0 , ϵ and T . $\epsilon - \epsilon_\theta(x_T, T)$ which ϵ_θ depends on x_T and T and the square which is basically your mean squared error. And this is what? This is my true noise. This is my predicted noise and the expectation value over all x_0 over x_0 , ϵ and

[28:41] T , okay? So, to train the model so it performs well on average across all images, okay? And all noise levels and time steps, okay? So, that is all images. So, this is x_0 , time step T , and ϵ is at all noise levels. So, during the training of diffusion model, this is the loss function. This is which is basically a mean squared error function. It is the difference between the true noise and the predicted noise, square of that one, absolute value, the square of that one. And then it is the and we take the expectation value over all images and all time steps over all noise levels, okay? So, with this one, I come to the end of today's session. Before I conclude, let us quickly summarize what

[29:33] we have learned. All throughout the discussions, we have we have gone through a lot of mathematical steps, but the main idea behind these mathematical steps are is that we already know what is the distribution, how the forward process, forward diffusion process distribution looks like. And using that information, we have modeled the reverse or the backward diffusion process in the form of a Gaussian distribution with certain mean and covariance, okay? Now, what we want, we want our neural network to learn the mean and the covariance of this Gaussian distribution.

[30:16] And for that one, once again, we have gone back to the equation, some mathematical calculations, mathematical, you know, steps that you know, we have undergone through. And then finally, we have found out the mean of the Gaussian distribution, which is followed in the reverse process, diffusion process. And that can

be written as only as a function of X_T and T at a particular time instant T without having any dependence on X_0 .

[30:51] Okay. So, once you know what Suppose you are at the time step T , small T , and you know what is the mean and the covariance which is dependent only on X_T and the time step T . Once you know that one, then you can generate or sample X_{T-1} which is your next step in the reverse trajectory. Okay. Once you can sample X_{T-1} given the mean and the covariance, how we will do that one? We have We know any sample can be uh you know in in in any step can be or any image step can be sampled from the Gaussian distribution which is equal to if it is X_{T-1} , it can be written as the mean plus Z into the variant the standard deviation. Okay. So, using that one, we have generated X_{T-1}

[31:47] one. Now, once that X_{T-1} is known to us, the neural network will learn the mean and the variance at the time X_{T-1} . Using that mean and the variance, it will generate X_{T-2} and it will keep on going on iteratively till we come to X_0 which is our clean image. Okay. So, with this one, I conclude today's session. I hope you have got an understanding of how the reverse diffusion process uh looks like. What are the mathematical steps that are involved in the reverse diffusion process?

[32:28] Okay. And uh in my next session, we will discuss about the unit architecture for denoising. So, what is the architecture that is followed for this denoising process? So, that we will look into in my next session. I hope this session is fruitful for you all. Till we meet for the next session, thank you all and happy learning.